

Industrial Augmented Reality – Feasibility of AI-based Validation of AR Instructed Component Recovery for Washing Machines in Circular Economy

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ABSTRACT

Using Artificial Intelligence (AI) to validate that tasks are performed correctly when instructed through Augmented Reality (AR) is a rather unexplored area. Focusing on the removal of components during the disassembly of washing machines in the context of circular economy, we develop an AR system, to detect correctly executed tasks using AI. We present preliminary results of our novel approach using the YOLOv8 Nano object model detecting eight components. Already with very limited training data we could achieve detection precisions of over 90% for seven of eight components proving the general feasibility of our work.

Keywords: Augmented Reality, Artificial Intelligence, Circular Economy, Disassembly, Validation.

Index Terms: Human-centered computing → Mixed / augmented reality; Computing methodologies → Artificial intelligence

1 INTRODUCTION

Augmented Reality (AR) has found its way into many application areas in industry, supporting design, maintenance and assembly. With the change from a linear production economy to a circular one, a new field with novel challenges for AR support systems arises: the disassembly process. Disassembly is the key process in circular economy where used components are recovered from used products in order to reuse them. The key difference to the assembly process is, that standardized products become individual products when they are returned for disassembly. At this stage, the same components within the different products are in different conditions for reuse. In the EU-funded project DiCiM (www.dicimproject.eu) we are developing a support system for the component recovery of washing machines, focusing on the problem that each returned machine needs to be treated individually at disassembly. One machine for example returns with a broken motor that needs to be scrapped whereas the next machine of the same type might contain a directly reusable motor and the motor of a third machine has to be refurbished before it can be reused. An AR system supporting this process therefore has to adapt the instructions individually for each machine. Another challenge when using AR instructions, independent of the supported process, is that it is never assured, that the task was carried out correctly. For the use case of our monitor-based AR disassembly system, we therefore aim to integrate an artificial intelligence (AI) based validation system, that confirms if

an instruction was carried out correctly. In the following, we present our approach and preliminary results of our AI validation system, which we find promising also in other application areas. The associated AR system is currently developed, meaning that the AI validation system is not yet integrated into the AR system.

2 STATE OF THE ART

Several object detection and tracking algorithms have been developed, primarily based on CNNs (convolutional neural network), e.g., SSD [1], and YOLO [2]. However, no literature was found on detecting washing machine parts during disassembly or assembly. The closest application from Tan et al. [3] utilize YOLOv5 for sorting washing machine waste using sensor data, achieving mean Average Precision (mAP) of 96.3% and 98.7%. Nowakowski et al. use CNNs and R-CNNs to detect and classify the size of e-waste items, like washing machines, refrigerators, and computer monitors, achieving accuracies between 90% and 97% [4]. Though, no parts were detected only the entire appliance. Zhang et al. [5] detect screws in consumer electronics using YOLOv4 and recommend tools for their disassembly after classifying screw types with EfficientNetv2. Regarding the detection of components inside of devices, Sterkens et al. identified and sorted batteries in 532 electronic devices using deep learning object detection with X-Ray images, achieving 89% precision and 81% recall [6]. Combining AI and AR has seen many applications in recent years [7], but predominately for enhancing AR technology itself and not for task validation. So far only few studies consider the automated validation of AR instructed tasks. Manuri et al. [8] and Alves et al. [9] use a computer vision based approach for task validation in a maintenance task for a laser calibrator and with LEGO building block models. Only Rheault et al. [10] utilize AI methods for guiding a user through a cooking task using AR.

3 AI VALIDATION METHOD CONCEPT

To address requirements for training CNNs, we used a transfer learning approach with YOLOv8 Nano [11]. This leverages pre-trained models, fine-tuning them with smaller, task-specific datasets. YOLOv8 Nano's streamlined CNN has fewer parameters, making it suitable for real-time tracking. Component detection involves a CNN assigning a class probability score via a Softmax function. Tracking provides several bounding boxes around an object, each with a confidence score based on factors like color similarity and adjacent cell shape. The overall score is the product of confidence and class probability scores. For a smooth detection, redundant bounding boxes are removed using Intersection over Union for non-maximum suppression at a threshold of 0.5. Therefore, bounding boxes with the highest scores are selected, eliminating overlaps and achieving precise localization, resulting in a single bounding box with its class probability for each detected object.

4 PRELIMINARY MODEL IMPLEMENTATION AND RESULTS

From one operator-recorded video (720x1280 pixels) of one washing machine disassembly, we extracted training images with two

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frames per seconds (450 images). We labeled these images to identify eight washing machine components, creating a dataset: Counter Weight Upper, Soap Dispenser Housing, RFI-Filter, Supply Hose, Control Unit, Control Knob, Control Panel and Hose ADS (connector valve/soap dispenser). This dataset was divided into training, validation, and testing subsets in a 70:20:10 ratio. After the first results we trained the model with 315 additional images. To enhance dataset robustness and diversity, we applied augmentation techniques via a two-step pipeline. Augmentations included horizontal flipping, adjusting brightness and contrast to simulate different lighting, and applying motion blur to mimic motion during assembly. These augmentations help the model to generalize to real-world scenarios. The YOLOv8 model was trained for 100 epochs with a batch size of 8 to detect the components. The feature map is processed by the detection head, which predicts bounding boxes and class probabilities for objects (Figure 1). Figure 2 shows the Precision-Recall curve of our model's performance across the eight components. Counter Weight Upper, Soap Dispenser Housing, RFI-Filter, Supply Hose, and Control Unit exhibit very high precision and recall (~97+%), indicating very accurate detections. Control Knob and Control Panel have mAPs of 90.6% and 92.4%, indicating reliable detections, Hose ADS has a lower mAP of 56.3%, highlighting detection challenges.

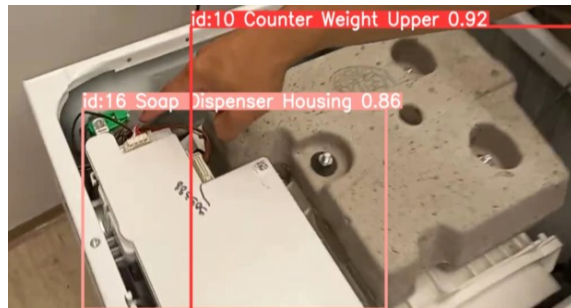


Figure 1: Detected component together with its score.

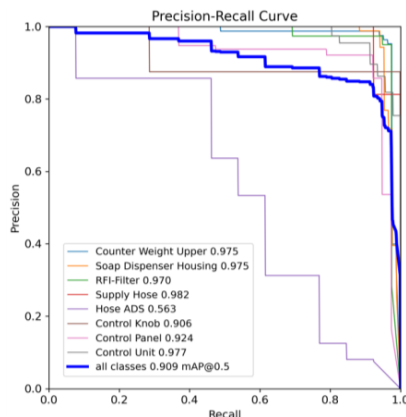


Figure 2: Precision-Recall curve of 8 components with their scores.

5 DISCUSSION AND FUTURE WORK

Despite the limited training dataset, the model achieved a high mAP score of 90.9% at a confidence threshold of 0.5. This indicates, the model performs well across all classes, accurately detecting most components, making it a solid choice for our application. The poor performance for detecting the Hose ADS component is due to underrepresentation in the training data relative to the other components. To overcome this, additional videos have been recorded to provide more balanced data for training, aiming to reduce bias and improve detection accuracy. The next step involves collecting more

images and evaluating depth images as a possible alternative to 2D images. Due to the limited training data, we are considering generating synthetic data from different angles based on 3D models and training the model along with the real image data.

6 CONCLUSION

We presented a novel detection approach for component recovering during washing machine disassembly in a circular economy context. Using our novel approach relying on the YOLOv8 model we could achieve > 90% detection rate for seven of the eight targeted components with few training data. These preliminary results show the feasibility of our approach to integrating AI task validation capabilities in AR support systems. Future work will refine the detection quality and the integration of the AI detection into the AR support application.

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